

Project Title
Email Content Analysis for Detecting Spam Messages

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Email Content Analysis for Detecting Spam Messages is a Java-based console application developed as an attempt to solve a common problem faced by email users: identifying unwanted and suspicious messages. The project focuses on analyzing the content of an email and finding words or phrases that may indicate spam, such as prize-related messages, account verification requests, password requests, or suspicious links. The email data and spam keywords are stored separately in text files, so new emails and keywords can be added easily without making changes to the main program. Before analysis, the email content is cleaned and processed so that the algorithms can search it more effectively. Different stringmatching techniques, including KMP, Rabin-Karp, Z-Algorithm, and AhoCorasick, are used to identify suspicious patterns in the email.

After finding the suspicious patterns, the project uses other DSA concepts to continue the analysis. Bitmask Dynamic Programming is used to calculate a feature score based on the detected patterns, while Edmonds-Karp is used to demonstrate how a network-flow approach can be applied to the detected spam features. A Greedy Set Cover method is used to select suitable rules that cover the suspicious patterns, and randomized hashing is included to demonstrate the use of randomization. The project also uses parallel processing so that multiple selected emails can be processed using different threads. Finally, the calculated score is compared with a predefined threshold and the email is classified as SPAM or NOT SPAM. The system displays the patterns found, algorithm results, classification reason, and a final summary of the analyzed emails. Overall, this project provides a practical way to understand how the different DSA concepts learned in the course can be combined and applied to a real-world problem such as email spam analysis.

Signature of the Guide